

Exercise 3: Individual Report

1. Project Context

This project investigates writer identification and writer retrieval on historical handwriting images. The goal is to represent each document image with a global descriptor and then retrieve documents written by the same writer. The required pipeline used SIFT descriptors, VLAD encoding, normalization, PCA whitening, and Exemplar-SVM. The individual work extends this baseline with additional experiments to understand which design choices improve or reduce retrieval performance.

2. Suggested Improvement Directions and Selected Work

The exercise allowed additional experiments beyond the required pipeline. Possible directions included PCA whitening, different VLAD/codebook configurations, learned CNN/ViT features such as DINOv2, input preprocessing changes, and other retrieval improvements. I selected three controlled experiments that are practical, measurable, and directly connected to the pipeline:

1. Experiment 1: PCA dimensionality sweep for the SIFT + VLAD representation.
2. Experiment 2: Codebook-size sweep for the VLAD representation.
3. Experiment 3: Learned feature experiment using DINOv2 patch descriptors, including an input-resolution comparison.

3. Required Baseline Used for Comparison

The following table shows the required group/baseline pipeline. This baseline is important because the individual experiments should be interpreted relative to a strong SIFT + VLAD result.

Method	Top-1	mAP
Plain VLAD	0.820556	0.629558
VLAD + power normalization	0.832500	0.644012
VLAD + power + intra-normalization	0.838889	0.649681
VLAD + power + intra + PCA whitening (512)	0.882778	0.744912
Exemplar-SVM on VLAD representation	0.895556	0.769657

The final required result was Exemplar-SVM with Top-1 0.895556 and mAP 0.769657. For the individual parameter sweeps, the cleanest comparison baseline is the PCA-whitened SIFT + VLAD descriptor with K=100 and PCA=512, which achieved Top-1 0.882778 and mAP 0.744912.

4. Experiment 1: PCA Dimensionality Sweep

4.1 Motivation

PCA whitening produced the largest improvement in the required pipeline, so I tested whether changing the number of PCA components would further improve performance. A smaller number of components may remove noise but can also remove useful writer-specific information. A larger number of components preserves more information but may also keep noisy or bursty dimensions.

4.2 Setup

- Fixed codebook size: K=100.
- Compared PCA components: 256, 512, and 1024.
- Compared row: powernorm_intranorm_vlad_pca_whiten.

4.3 Results

Experiment	Setting	Top-1	mAP	mAP change
PCA compression	K=100, PCA=256	0.877778	0.737041	-0.007871
Baseline	K=100, PCA=512	0.882778	0.744912	+0.000000
More components	K=100, PCA=1024	0.879444	0.745972	+0.001061

4.4 Interpretation

Using only 256 PCA components decreased mAP from 0.744912 to 0.737041. This means the representation was compressed too strongly and probably lost useful handwriting information.

Increasing the number of PCA components from 512 to 1024 gave only a very small improvement, from 0.744912 to 0.745972 mAP. This suggests that PCA=512 already preserved most of the useful information in the K=100 VLAD representation.

5. Experiment 2: Codebook-Size Sweep

5.1 Motivation

The codebook size K controls the number of visual words used by VLAD. A smaller K gives a compact descriptor but may merge different handwriting patterns. A larger K can represent more detailed local structures, but it also increases descriptor dimensionality and sparsity. Therefore, I tested whether changing K improves retrieval.

5.2 Setup

- Compared codebook sizes: K=64, K=100, and K=256.
- K=100 PCA=512 was used as the main baseline.
- For K=256, both PCA=512 and PCA=1024 were evaluated.

5.3 Results

Experiment	Setting	Top-1	mAP	mAP change
Smaller codebook	K=64, PCA=512	0.879167	0.739265	-0.005647
Baseline	K=100, PCA=512	0.882778	0.744912	+0.000000
Larger codebook	K=256, PCA=512	0.881111	0.746333	+0.001421
Larger codebook + more PCA	K=256, PCA=1024	0.883611	0.749942	+0.005030

5.4 Interpretation

Reducing the codebook to K=64 decreased mAP to 0.739265. This suggests that the smaller vocabulary was too compact and lost fine-grained stroke information.

Increasing the codebook to K=256 with PCA=512 gave mAP 0.746333, a small improvement over the K=100 PCA=512 baseline. Increasing PCA components to 1024 for K=256 improved the result further to mAP 0.749942. This was the best SIFT-based individual result. However, the improvement over the original K=100 PCA=512 baseline was only +0.005030, so the gain is real but modest.

6. Experiment 3: Learned Feature Experiment with DINOv2

6.1 Motivation

The third experiment replaced handcrafted SIFT descriptors with learned DINOv2 patch-token descriptors. The purpose was to compare a classical local descriptor with a modern self-supervised ViT feature extractor. The rest of the pipeline was kept similar: descriptors were clustered, encoded with VLAD, normalized, whitened with PCA, and evaluated with Top-1 accuracy and mAP.

6.2 Setup

- Model: DINOv2 ViT-S/14 patch-token descriptors.
- Input preprocessing: inverted binary images.
- Codebook size: K=64.
- Maximum dictionary descriptors: 100000.
- PCA whitening components: 512.
- Compared image sizes: 224 and 336.

6.3 Results

Experiment	Setting	Top-1	mAP
DINOv2 low resolution	Image size=224, K=64, PCA=512	0.233611	0.145781
DINOv2 higher resolution	Image size=336, K=64, PCA=512	0.331111	0.202744

6.4 Resolution Comparison

Increasing DINOv2 input resolution from 224 to 336 improved Top-1 by +0.097500 and mAP by +0.056963. This shows that DINOv2 benefits from preserving more spatial handwriting detail. Full document images resized to 224 pixels lose many small stroke structures, while 336 pixels keeps more useful local information.

6.5 Comparison with SIFT

The best DINOv2 result was image size 336 with mAP 0.202744. This did not outperform the SIFT + VLAD + PCA512 baseline with mAP 0.744912, nor the final SIFT + Exemplar-SVM result with mAP 0.769657. The difference between best DINOv2 and SIFT + VLAD + PCA512 was -0.542168 mAP.

A likely reason is domain mismatch: DINOv2 is a general self-supervised visual feature extractor, while the dataset consists of binarized historical handwriting. SIFT directly captures local gradient and stroke structures, which are very relevant for writer identification. DINOv2 may require text-region cropping, higher input resolution, a different codebook, or task-specific tuning to become competitive in this pipeline.

7. Conclusion

The individual experiments provide three main findings. First, PCA=512 is already a strong compression setting for K=100; PCA=256 loses information, while PCA=1024 only gives a tiny gain. Second, increasing the codebook size to K=256 gives the best SIFT-based individual result when combined with PCA=1024, reaching mAP 0.749942. Third, DINOv2 did not outperform the SIFT pipeline, but increasing input resolution from 224 to 336 clearly improved DINOv2 performance, showing the importance of fine stroke-level detail.

The strongest individual result was K=256 with PCA=1024 in the SIFT + VLAD pipeline. The most conceptually different experiment was DINOv2, which provided a useful comparison between handcrafted and learned features even though it did not improve the final score.

Appendix A: Raw Result Tables

This appendix includes all uploaded result rows used in the report. The small DINO test is included separately as a sanity check and was not used for the final full-dataset comparison.

A.1 Required baseline: SIFT + VLAD, K=100, PCA512, E-SVM

Result name	Top-1	mAP
plain_vlad	0.820556	0.629558
powernorm_vlad	0.832500	0.644012
powernorm_intranorm_vlad	0.838889	0.649681
powernorm_intranorm_vlad_pca_whiten	0.882778	0.744912
esvm_powernorm_intranorm_vlad	0.895556	0.769657

A.2 PCA sweep: K=100, PCA256

Result name	Top-1	mAP
plain_vlad	0.820556	0.629558
powernorm_vlad	0.832500	0.644012
powernorm_intranorm_vlad	0.838889	0.649681
powernorm_intranorm_vlad_pca_whiten	0.877778	0.737041

A.3 PCA sweep: K=100, PCA1024

Result name	Top-1	mAP
plain_vlad	0.820556	0.629558
powernorm_vlad	0.832500	0.644012
powernorm_intranorm_vlad	0.838889	0.649681
powernorm_intranorm_vlad_pca_whiten	0.879444	0.745972

A.4 Codebook sweep: K=64, PCA512

Result name	Top-1	mAP
plain_vlad	0.811389	0.616268
powernorm_vlad	0.822222	0.631309
powernorm_intranorm_vlad	0.826944	0.636630

powernorm_intranorm_vlad_pca_whiten	0.879167	0.739265
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A.5 Codebook sweep: K=256, PCA512

Result name	Top-1	mAP
plain_vlad	0.833333	0.649783
powernorm_vlad	0.839722	0.659882
powernorm_intranorm_vlad	0.846389	0.663090
powernorm_intranorm_vlad_pca_whiten	0.881111	0.746333

A.6 Codebook sweep: K=256, PCA1024

Result name	Top-1	mAP
plain_vlad	0.833333	0.649783
powernorm_vlad	0.839722	0.659882
powernorm_intranorm_vlad	0.846389	0.663090
powernorm_intranorm_vlad_pca_whiten	0.883611	0.749942

A.7 DINOv2 full dataset: image size 224, K64, PCA512

Result name	Top-1	mAP
dino_plain_vlad	0.219167	0.136977
dino_powernorm_vlad	0.234722	0.142033
dino_powernorm_intranorm_vlad	0.215556	0.130189
dino_powernorm_intranorm_vlad_pca_whiten	0.233611	0.145781

A.8 DINOv2 full dataset: image size 336, K64, PCA512

Result name	Top-1	mAP
dino_plain_vlad	0.307500	0.184858
dino_powernorm_vlad	0.307222	0.184557
dino_powernorm_intranorm_vlad	0.290833	0.171793
dino_powernorm_intranorm_vlad_pca_whiten	0.331111	0.202744

A.9 DINOv2 small sanity test: image size 224, K32, PCA512

Result name	Top-1	mAP
dino_plain_vlad	0.470000	0.334511
dino_powernorm_vlad	0.450000	0.332045
dino_powernorm_intranorm_vlad	0.420000	0.318008
dino_powernorm_intranorm_vlad_pca_whiten	0.100000	0.099340